Managerial Economics

Lecture 6 Firm Organization Analysis

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RENMIN BUSINESS SCHOOL



Opening Case Study: Xiaomi's Breakthrough Strategy

2015 marked a turning point in Xiaomi‘s development. With the proliferation of smartphones, the Chinese mobile phone market began to saturate, and competition on low prices became increasingly fierce. In 2016, Xiaomi’s mobile phone sales dropped by 36%, and the company‘s development encountered a deadlock... After analyzing the situation, Lei Jun believed that smart hardware interconnected in the era of the *Internet of Things (物联网)* is an inevitable trend, and Xiaomi needed to enrich its product content and variety. A series of questions were laid out in front of Lei Jun:

- Which products should Xiaomi expand into? Should they be related or unrelated to the core product?
- How should Xiaomi expand into other products? Should they **produce on their own, outsource,** or **form joint ventures**?

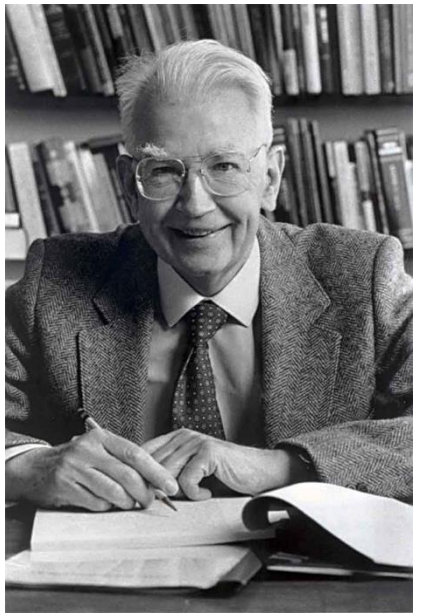
Learning Goals

- How to Minimize the Cost of Factor Inputs Using Different Methods
- Four Types of Specificity Investments and the Problems They Cause
- Choosing the Best Procurement Method for Different Types of Factor Inputs
- Principal-Agent Problems and Incentive Mechanism Design

The Optimal Way to Acquire Production Inputs

The Nature of Firm

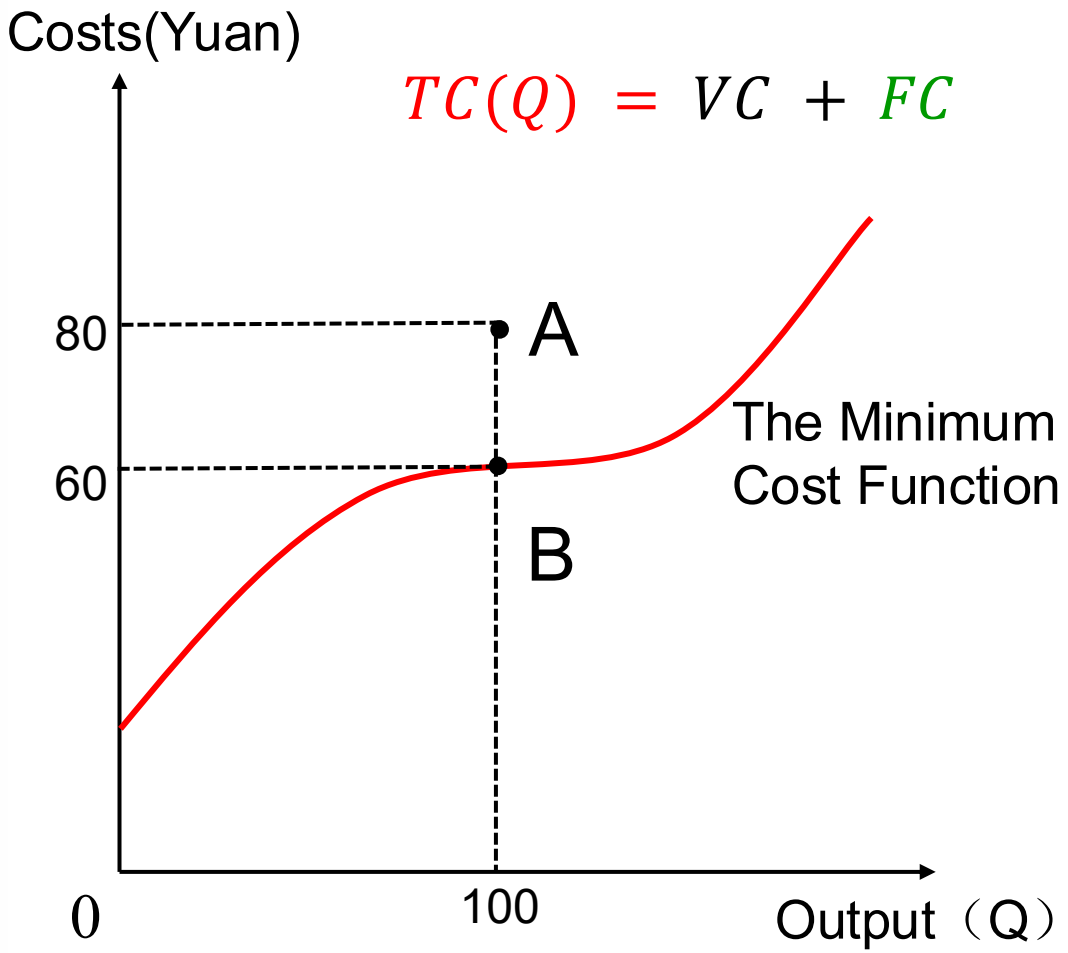
- **Neoclassical Economics:** Values market and price mechanisms, emphasizes production efficiency and profit maximization, and views firms as part of the production function.
- **Coase's Question (The Nature of the Firm, 1937):** Why do firms exist in the market instead of all production activities being carried out through market transactions?
  - **Coase's Explanation:** Market transactions involve transaction costs.
  - The relative size of internal transaction costs within a firm compared to market transaction costs determines the boundaries of the firm.
- **New Institutional Economics:** Focuses on the *internal organizational structure* and *governance* of firms, emphasizes *transaction costs* and *contractual arrangements*
  - Highlights the role of *legal*, *policy*, and cultural backgrounds in the formation of firms.



Ronald Coase
(罗纳德•科斯)
Economist
1910-2013

More Production Costs Definitions

- The cost for a company to produce 100 units of output at point A is 80 yuan, which is higher than the minimum cost of 60 yuan at that point.
- The company has not achieved the optimal combination of inputs, or there is excessive slacking among employees.
- How should the company achieve the optimal combination of input factors to minimize costs?



Three Basic Ways to Acquire Production Inputs

1. **Spot Exchange (现货市场交易)**: The buyer and seller meet, trade, and then part ways without any contract.
   - Managers choose to purchase production inputs from one of many supplies in the market
   - **Pros:** Clear division of labor between the enterprise and producers, each focusing on their respective strengths.
   - ✗ **Cons:** Uncertainty in the buy-sell relationship, quality management.
2. **Outsourcing Through Contract (合同外包）:** Establishing a cooperative relationship between trading parties through a contract.
   - The contract specifies the scope of services, prices, and timing
   - **Pros:** Clear division of labor between the enterprise and producers, stable cooperative relationship, clear constraints within the contract.
   - ✗ **Cons:** Higher costs (drafting fees, legal fees), contract incompleteness

Three Basic Ways to Acquire Production Inputs

3. Vertical Integration (纵向一体化): Bypassing other suppliers and producing inputs within the company

- The enterprise produces both the inputs and the final products.
- **Pros:** Independent supply chain, less dependency on suppliers: good quality control.
- ✗ **Cons:** Detrimental to the enterprise's specialized division of labor advantage, increases organizational costs.

Quiz: Determine the Method of Acquiring Inputs

1. In accordance with the agreement, Company A will purchase 1,000 computer chips from Company B each year for the next three years. The price for the first year is 1,500 yuan per chip, with the price in the second and third years increasing proportionally according to the annual wholesale price index of chips.

   Contract

2. Company A1 purchased 500 chips from Company B1 through the Alibaba platform.

   Spot Transaction

3. Company A2 acquired Company B2, a chip manufacturer, to produce motherboards and chips for its own computers.

   Vertical Integration

Transaction Costs（交易成本）

- **Transaction Costs**: The part of the cost incurred by a company in acquiring inputs that exceeds the actual payment made to the supplier.
- Transaction costs are incurred in the entire transaction process:
  - The cost of searching for established input suppliers.
  - The cost of negotiating inputs, such as time invested, legal fees, etc.
  - Investments or expenditures related to facilitating the transaction.
- **Explicit transaction costs**, e.g., transportation and logistics costs.
- **Hidden transaction costs,** e.g., routine and specific to the transaction.
- A transaction is a relationship-specific exchange (关系专用性交易) if it requires specialized investment(专用性投资) to facilitate.

Types of Specialized Investment

- **Site Specificity (地点专用性):** Choosing a nearby location for a factory to reduce transaction costs.
  - Power plants are often built near coal mines.
- **Physical Asset Specificity (实物资产专用性)**: Machinery and equipment used to produce a certain input cannot produce other inputs.
  - The Equipment for producing smartphone screens cannot be used to produce smartphone chips.

Types of Specialized Investment

- **Specialized Assets (专用性资产)**: General investments made by a company for transactions with certain important customers.
  - A computer manufacturer investing in a new production line specifically to meet the demands of a large government purchaser.

- **Specialized Human Capital(专用性人力资本)**: Specific skills that employees must learn to work in a certain company, which are of no value or non-transferable to other employers.
  - Such as workplace relationships, minor service tasks.

Three Implications of Specialized Investment

1. High Bargaining Costs

- *Low transaction costs, consistent input quality, and multiple suppliers*
  - Market supply and demand forces will determine the appropriate price.
  - Electronic procurement platforms reduce search costs and increase price transparency.
- **Specialized investment:** few suppliers, and buyers need to make specialized investments, without market price for reference.
  - Prices are determined by negotiations.
  - E.g., market transactions of patents, and horizontal cooperative research between enterprises and universities.
- **High negotiating costs**: both parties often need to hire representatives to negotiate for favorable prices.
  - Negotiation skills and strategic games determine the final price.

2. Underinvestment in the Specialized Investment

- When transactions require specialized investments, the level of such investments often falls below the optimal level.
- Specialized investors will try to avoid being 'trapped' (套牢)by their trading partners, losing bargaining power:
  - University professors: *Research* vs. *Teaching*.
  - Employees intending to switch jobs lack enthusiasm for learning specialized skills at their current company.
  - Suppliers meet specific customer needs by purchasing relatively cheap machines.
- Patent investments often lead to **lower-quality products** and **higher transaction costs**.

3. Opportunism (机会主义） and Hold-up (敲竹杠) problem

- If acquiring inputs requires specialized investment, both the buyer and seller can exploit the **'sunk' nature** of the specialized investment for opportunistic behavior.
- **‘Hold-up’:** Once a company has made a specialized investment, the trading party might exploit the ‘sunk’ nature of this investment to **extort (敲诈)** the other party

Example: Opportunism and Hold-up Problem

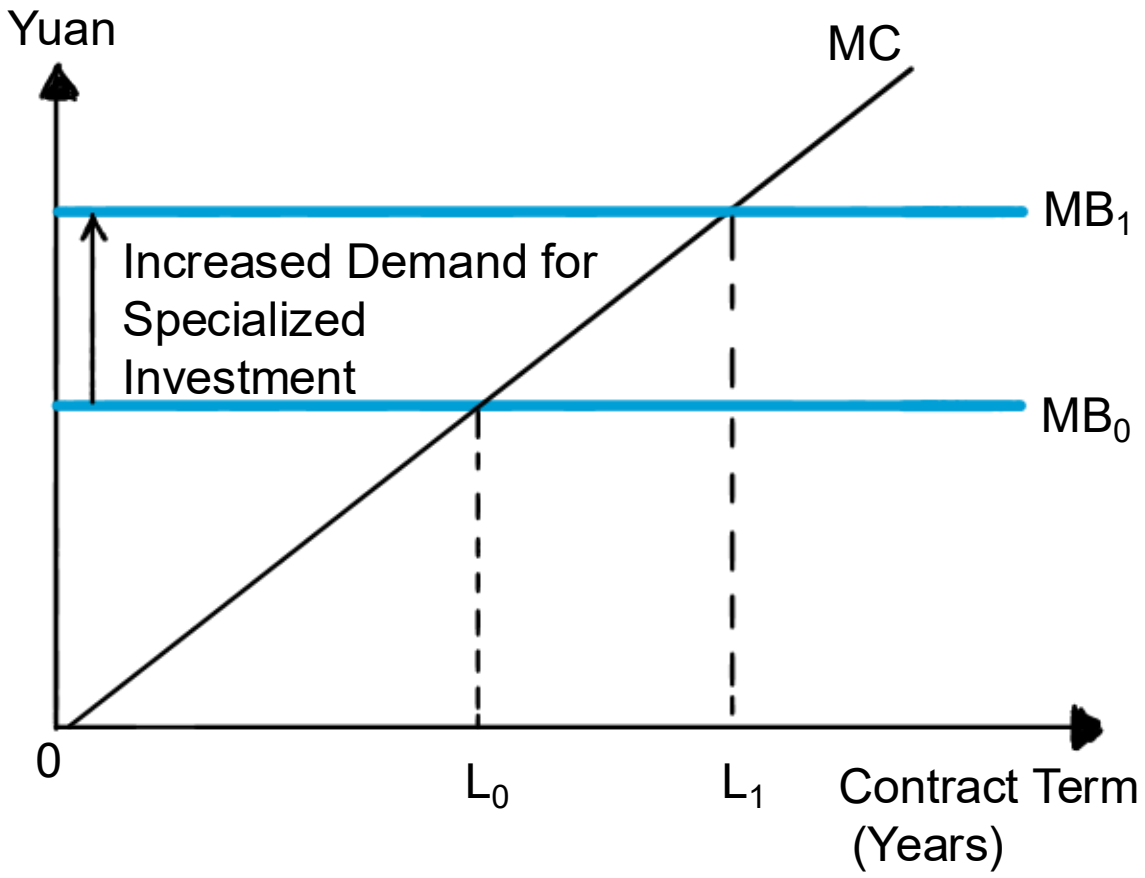
Example: Suppose a buyer needs to spend 500 yuan to quality-check a specific input. Multiple suppliers in the market are willing to sell this input for 5,000 yuan. The buyer randomly selects a supplier and spends 500 yuan to check the product quality. This scenario can lead to opportunistic behavior by the supplier.

- The supplier may raise the selling price to 5,499 yuan, noting that the buyer's cost of purchasing the input from other suppliers is at least 5,500 yuan. The buyer might end up having to purchase the input from this supplier

The Optimal Way to Acquire Inputs

- **Spot Transaction:** With the presence of specific investments, spot transactions may incur higher transaction costs due to *opportunism, high bargaining costs, and underinvestment*.
- **Contract**: Purchasing inputs from a specific supplier through a contract can prevent hold-up behavior and repetitive bargaining.
  - Defining the **quantity** and **price** of inputs *before making specialized investments* reduces opportunistic behavior.
  - The contract stipulates that both parties *adhere to the agreed prices and quantities over a longer period* to incentivize specialized investments.
  - The optimal **contract term (合同期限)** depends on the marginal benefits and costs of the specialized investment.

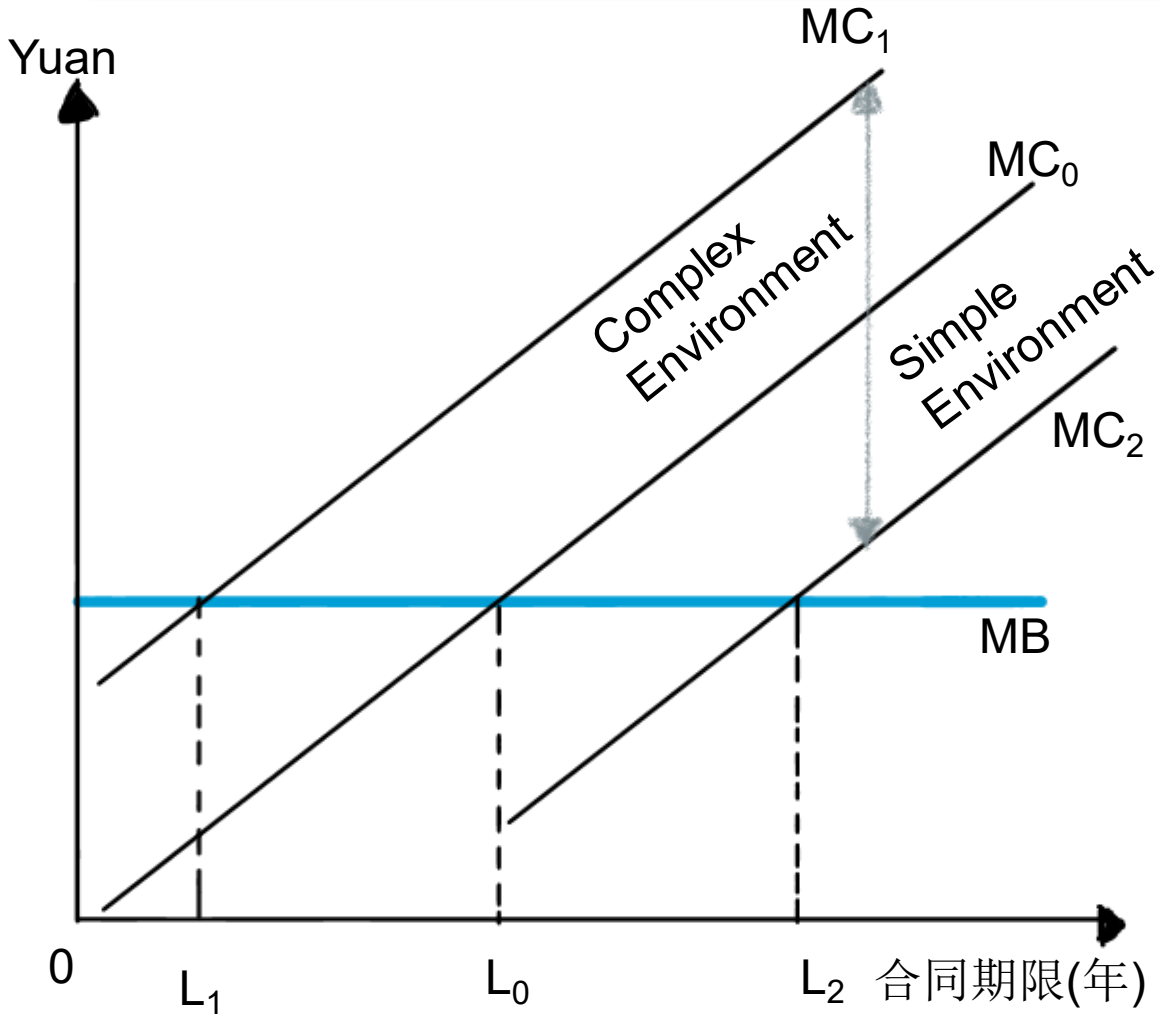
Specialized Investment and Contract Term(合同期限)



- When specialized investment becomes more critical, the transaction costs after the contract expires are higher. Signing a longer-term contract can avoid these costs, resulting in higher marginal benefits.

- As specialized investments increase, the optimal contract duration will also be extended.

Contract Environment and Contract Term(合同期限)

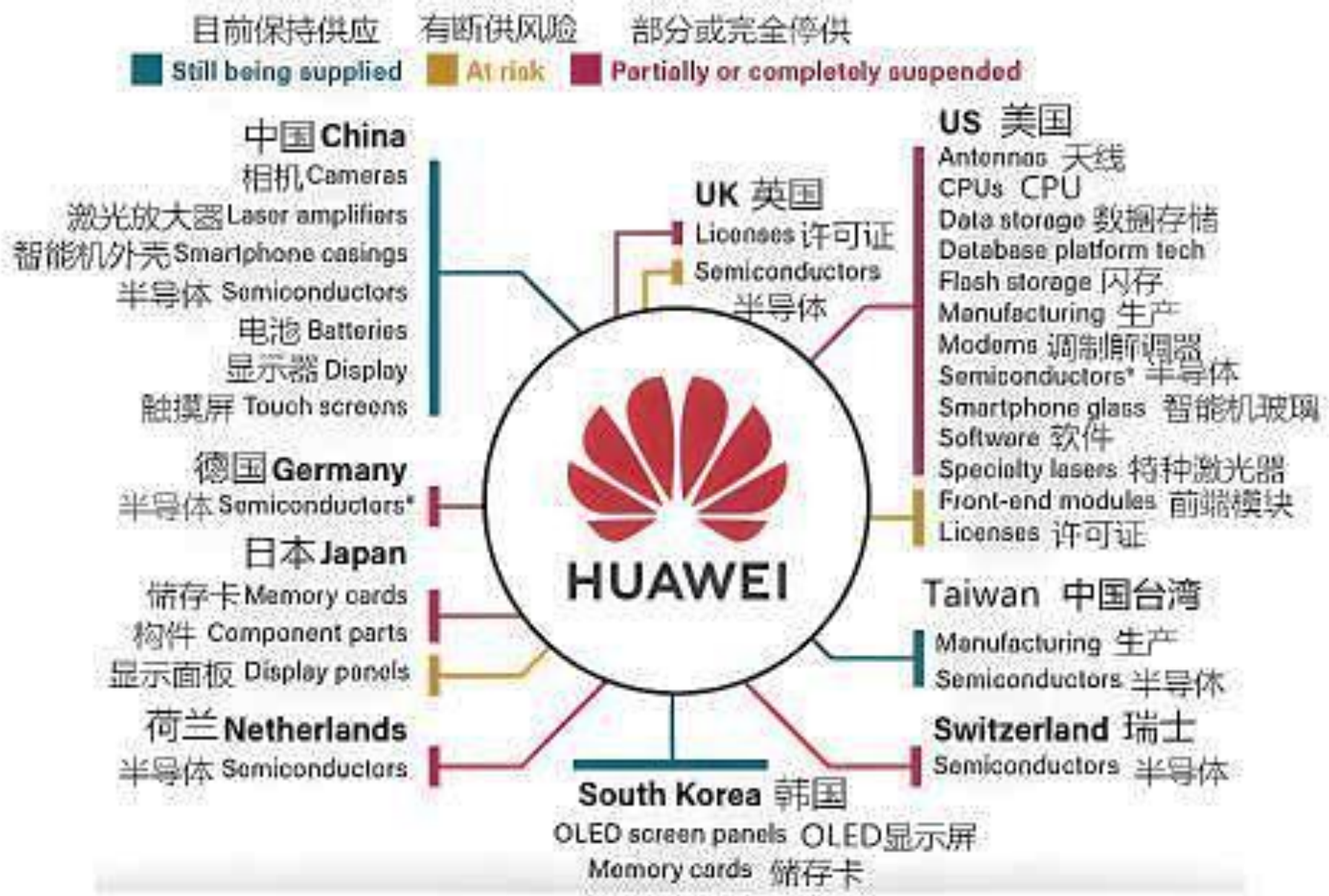


- When inputs are more *standardized* and the future economic environment is more *certain*, the marginal cost of signing a contract decreases, which will *extend* the optimal contract period

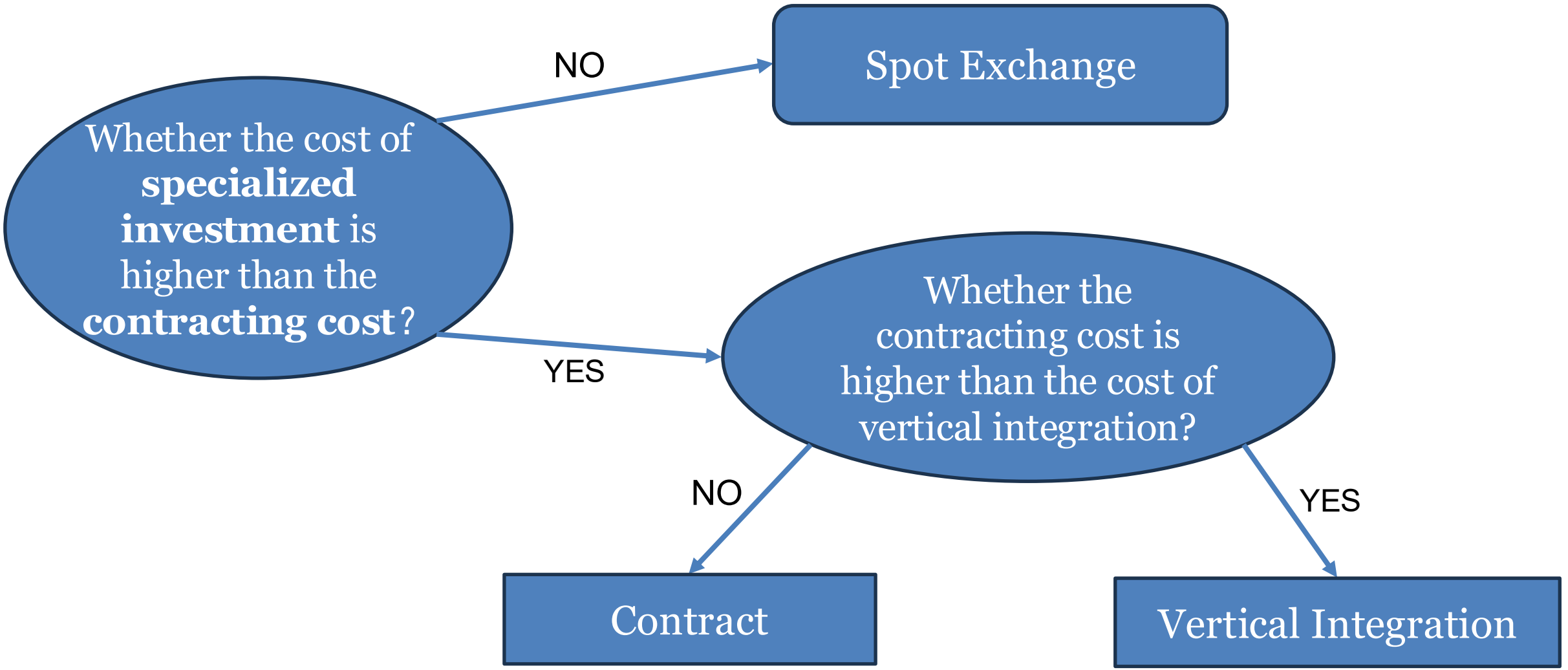
- When the production process of inputs becomes more *complex* and the future economic environment becomes more *uncertain*, the contract becomes more detailed, which *shortens* the optimal duration of the contract

Vertical Integration (纵向一体化)

- Managers can produce inputs themselves through vertical integration to reduce the transaction costs caused by market uncertainty
  - *Incomplete contracts* or *high costs of contract signing*
  - E.g., Automobile manufacturers produce their own baffles（挡板）; mobile phone manufacturers design and make their own cameras
- Advantages and disadvantages of vertical integration:
  - **Advantage:** Integrate specialized manufacturers into specialized companies, and reduce opportunistic behavior
    - Replacing market resource allocation with internal resource allocation methods to reduce transaction costs
  - ✗ **Disadvantage**: High costs of constructing the production line
- Vertical integration is often the last choice for enterprises


目前保持供应
Still being supplied
有断供风险
At risk
部分或完全停供
Partially or completely suspended
中国 China
相机 Cameras
激光放大器 Laser amplifiers
智能机外壳 Smartphone casings
半导体 Semiconductors
电池 Batteries
显示器 Display
触摸屏 Touch screens
德国 Germany
半导体 Semiconductors*
日本 Japan
储存卡 Memory cards
构件 Component parts
显示面板 Display panels
荷兰 Netherlands
半导体 Semiconductors
UK 英国
Licenses 许可证
Semiconductors
半导体
US 美国
Antennas 天线
CPUs CPU
Data storage 数据存储
Database platform tech
Flash storage 闪存
Manufacturing 生产
Modems 调制解调器
Semiconductors* 半导体
Smartphone glass 智能机玻璃
Software 软件
Specialty lasers 特种激光器
Front-end modules 前端模块
Licenses 许可证
HUAWEI
Taiwan 中国台湾
Manufacturing 生产
Semiconductors 半导体
Switzerland 瑞士
Semiconductors 半导体
South Korea 韩国
OLED screen panels OLED显示屏
Memory cards 储存卡

The Economic Trade-off in Choosing the Optimal Method of Acquiring Inputs



Short Case Study:
The contractual relationship between airlines and engine companies

Blue Sky Airlines has signed a purchase contract with Baiyun Engine Company, promising to purchase 50 jet engines at a price of $200,000 per unit in the future. Because the contract had already been signed, Blue Sky Airlines believed that it could avoid the opportunistic behavior of Baiyun Engine Company, so they began designing an aircraft body suitable for Baiyun Engine. Due to the uncertainty of the aviation engine market, Baiyun Engine Company was on the brink of bankruptcy in the second year, so it proposed to Blue Sky Airlines that it would choose bankruptcy unless the engine price was raised to $300,000 per unit.

- What should the managers of Blue Sky Airlines do?
- How can Blue Sky Airlines avoid the aforementioned issues from occurring?
- Is it wise for Blue Sky Airlines to choose to purchase inputs through contracts?

Principle-Agent Problem and Salary Management

The Principle-Agent Problem

- Two sufficient conditions for principal-agent problems:
  - Different goals between principal and agent
  - The principal finds it difficult to effectively monitor the agent

- The separation of ownership and management rights is one of the characteristics of large enterprises:
  - It is difficult for owners to perfectly supervise the daily operation of managers.
  - The fundamental cause of the principal-agent problem.

The Principle-Agent Problem

- Due to the lack of supervision over managers, fundamental incentive issues arise:
  - When a company incurs losses, managers can always defend themselves against the owners:

You should be glad to have hired me. If it weren't for me working 18 hours a day, the company's losses would have been even greater. I think the company's situation will improve next year...

- There is also a principal-agent problem between managers and employees
  - Managers are unable to supervise the work of each employee, and it is difficult to restrain their *idle chatter* and *wandering* during working hours

Incentivize Managers: Fixed Wages and Incentive Contracts

Table 1 Firm Profits and Manager's Income under Fixed Wage

Manager's Income (10,000 Yuan)	Working time (hours)	Leisure Time (hours)	Firm Profits (10,000 Yuan)
80	8	0	1000
80	5	3	800
80	2	6	500
80	0	8	0

Table 2 Firm Profits and Manager's Income under Profits Sharing

管理者分享利润（万元）	管理者工作时间（小时）	管理者摸鱼时间（小时）	企业利润（万元）
100	8	0	1000
80	5	3	800
50	2	6	500
0	0	8	0

How to Constrain Managers?

- **Incentive contract:** CEO's income is linked to company performance, such as equity and profit sharing.
  - **Question:** The income level and structure of CEOs in different companies are not the same. How to evaluate whether the compensation of a company's CEO is reasonable?
    If the majority of CEO's income comes from performance bonuses, reducing their compensation is inappropriate
- **External incentives**:
  ① **Reputation:** In the long run, reputation will enhance the bargaining power of managers in the manager market, and companies will compete to hire excellent managers.
  ② **Acquisition threat:** If the manager is not doing well, the company will be acquired and the manager may also be replaced.

Solutions to Manger-Worker Principle-Agent Problem

1. **Profit sharing (利润共享):** Linking employee compensation to the potential profitability of the enterprise and encouraging employees to work hard. E.g., eewards for outstanding employees, etc.

2. **Benefit sharing (收益共享):** Combining employee compensation with the potential benefits of the enterprise. Such as tips for restaurant waiters, sales commissions, etc
   - ✯ The revenue sharing plan may increase the cost of providing goods or services: restaurant waiters may provide consumers with more beverages and a larger share of food.

3. **Piece rate wage (计件工资) :** a method of calculating wages based on workload rather than fixed hourly rates. E.g., manufacturing industry pays wages based on the quantity of products.
   - ✯ Employees may sacrifice quality in pursuit of quantity

Solutions to Manger-Worker Principle-Agent Problem

4. Attendance clock in (出勤打卡): Many companies use attendance records to monitor employee work. etc

- ✯ It is ineffective in solving principal-agent problems because it cannot detect the actual level of effort.

5. On site surprise inspections (现场突击检查): Managers periodically enter the work site to supervise employees to verify whether they are present at the work site, their level of effort, and the quality of their work.

- ✯ Unlike performance bonuses that motivate employees through reward commitments, *on-site inspections motivate employees through threats*.

Discussion 1: The Payment Method for Secretry's Salary

Your boss has just obtained an MBA degree from a certain business school and is quite interested in the company's salary management issues. He asked you: Why do companies pay hourly wages to secretaries instead of piece rate wages or commission based on the company's profit percentage? Can you answer his question?

- The work of a secretary is more easily supervised, and the principal and agent are not truly separated
- If piece rate wages are implemented, secretaries will focus on the quantity of work rather than the quality of work, such as increasing the number of printed documents, adding office consumables such as toner cartridges, etc

Discussion 2: The Payment Method for Secretry's Salary

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Discussion 2: The Principle-Agent Problem in the Class of Managerial Economics

Professor Chen teaches MBA's Management Economics. In order to enable students to fully understand and master the theoretical knowledge and practical applications of Management Economics, he wants to motivate them to work hard to learn Management Economics. Due to the inability to perfectly monitor the learning situation of students, he faces a principal-agent problem when motivating them to learn.

- Please discuss the role of conventional course evaluation supervision methods in solving the principal-agent problem in the teaching of Management Economics separately.

Summary

- Reasonably selecting and combining *market transactions*, *contracts*, and *vertical integration* methods to achieve cost minimization of factor inputs
- Four types of specialized investments and the problems they cause
- Choose the best input method based on the intensity of proprietary investment and the contract environment
- The principal-agent problem and incentive mechanism design
  - Between firm owner and the manager
  - Between manager and workers